

Aravind Centre for Eye Care Innovation (ACEi)

- Driving Innovation for Better Eye Care

As per the words of Dr. V, “Much has been done, but much remains to be done....” Innovation has always been an integral part of Aravind, driving more and more improvements in patient care, service delivery, and operational excellence. Building on this legacy, the Aravind Centre for Eye Care Innovation (ACEi) has continued to strengthen its role as Aravind’s dedicated platform since 2018. It functions as a catalyst in nurturing ideas, translating clinical challenges into practical solutions, and fostering a culture of innovation across the organisation.

Infrastructure Development

Eyedeas Lab, a dedicated environment for ideation, rapid prototyping, product development, and collaborative innovation, was set up at LAICO building and inaugurated on 28th June 2025. Established under the Empowering ACEi Project, the lab significantly enhanced ACEi’s capacity to transform innovative concepts into affordable, scalable, and clinically relevant solutions. Preparations began for establishing the EYERIS Lab, the innovations lab, at Aravind-Chennai, technical staff underwent intensive one-month training at the Eyedeas Lab and Aurolab to strengthen in-house expertise in product design, engineering, and prototyping.

New Initiatives

So far, ACEi collaborated with nine universities and start-ups, developed 28 prototypes, and generated 12 research and innovation outputs presented or published. These initiatives reinforced the department’s mission of promoting frugal innovation while addressing real-world clinical and operational challenges.

Inauguration of the Eyedeas lab at Aravind-Madurai

On 5th February 2026, the Aravind Eye Care System signed a Memorandum of Understanding with the PolyU Research Centre for SHARP Vision (RCSV), Hong Kong, to jointly develop and clinically validate EyeAgent, an artificial intelligence-powered clinical decision-support platform for ophthalmology. This partnership represents an important step toward integrating advanced AI technologies into eye care.

Regulatory affairs

To strengthen institutional support for innovation and intellectual property, the ACEi-Intellectual Property Rights (IPR) Committee was constituted during





Eye Mount 360

March 2026. The committee will guide patenting, technology transfer, commercialisation, and protection of innovations developed across the Aravind Eye Care System (AECS).

From idea to reality (Product development)

Several innovative products advanced substantially during the year. Amongst, Eye Mount 360 was officially launched in collaboration with LVPEI BioNEST Incubator, during the 84th Annual Conference of the All India Ophthalmological Society (AIOS 2026) held in Jaipur on 12–15 March 2026, and a patent application was filed.

KeraScan, a portable keratoconus screening device, emerged as one of the flagship developments. Originally

KeraScan



conceived through a clinical collaboration led by Dr. Anitha Karthikeyan, Johns Hopkins University, the concept was transformed by Eyedeas Lab into a portable, low-cost, webcam-based Placido disc corneal topographer. Through engineering redesign, 3D printing, and rapid prototyping, the device became suitable for community-based and paediatric screening, enabling early detection of keratoconus in high-risk populations.

Another significant innovation was GonioGlobe, a structured gonioscopy training and simulation model. Recognising the challenges associated with learning this highly subjective clinical skill, ACEi developed an affordable silicone eye model with interchangeable iris-angle plates that simulate normal and pathological angle configurations. The model enables repeated, risk-free practice for residents, fellows, and glaucoma surgeons while standardising training for angle-based procedures.

ACEi also developed practical solutions to improve clinical workflow and patient care. The A-Scan Probe Holder, designed using 3D printing technology, addressed operational issues by securely holding the probe after use while preventing cable interference and accidental damage. The LIDORA Eye Ointment Applicator was designed to help elderly patients, children, and individuals with limited dexterity apply ophthalmic ointments safely, accurately, and hygienically. Prototype refinements also continued for the 3D-Printed Prosthetic Eye, incorporating improved iris printing techniques, enhanced colouring, and transparent resin coatings to achieve a more realistic appearance.

Operational innovation remained another important focus area. Improvements to the Water Management System included the development of an ultrasonic flow-meter-based monitoring system with dashboard integration, enabling accurate water measurement, enhanced traceability, and improved system reliability. Similarly, the Eyeball Holder underwent design modifications to provide improved vacuum stability and enhanced safety during eye bank procedures and surgical training. Development also progressed on the Slit Lamp Sensor, a LiDAR-based plug-and-play device that automatically controls slit lamp illumination based on clinician presence, improving energy efficiency and workflow without requiring modifications to existing equipment.

A technical visit to Chemzest Silicone Company, Chennai, further strengthened product development efforts by identifying suitable silicone materials for



The Hackathon

ophthalmic simulation models, including GonioGlobe and the Squint Surgery Simulation Model.

Events

A two-week training programme on 3D Design and Printing was conducted from 2-13 September 2025 for six Instrumentation (Biomedical Engineering) interns, providing hands-on experience in 3D design, printing, and printer operation. The training covered the fundamentals of 3D printer operation, printer usage and maintenance, the PrusaSlicer application, CAD design using Autodesk Fusion 360, and the use of Vernier calipers. The programme supports the vision of establishing Eyedeas Labs across all Aravind Eye Hospitals by strengthening in-house capabilities in 3D design, rapid prototyping, and innovation.

The Hackathon 2026, conducted at Aravind-Pondicherry from 2-4 January 2026, brought together doctors, engineers, and students to solve clinical, surgical, patient care, and training challenges. From 169



3D design, rapid prototyping

problem statements, 27 challenges were shortlisted, attracting 260 applications from nine institutions. Winning teams received prizes of up to Rs. 50,000, while several promising ideas progressed toward prototype development through mentorship from ACEi.

Aravind Inspiring Innovator Award (AIIA)

The culture of innovation was further strengthened through the Aravind Inspiring Innovator Award (AIIA) 2026, which recognised outstanding innovations from across the organisation. . At the awards ceremony held at Aravind-Pondicherry on xxx, the Time Lapse Tomography (TLT) project received the top award, while Intraoperative Fluorescein Angiography (IOFA) secured the first runner-up position. LIDORA and Lazeyetrack were jointly recognised as second runner-up innovations, highlighting the diversity and quality of innovations emerging from clinicians and trainees across Aravind.